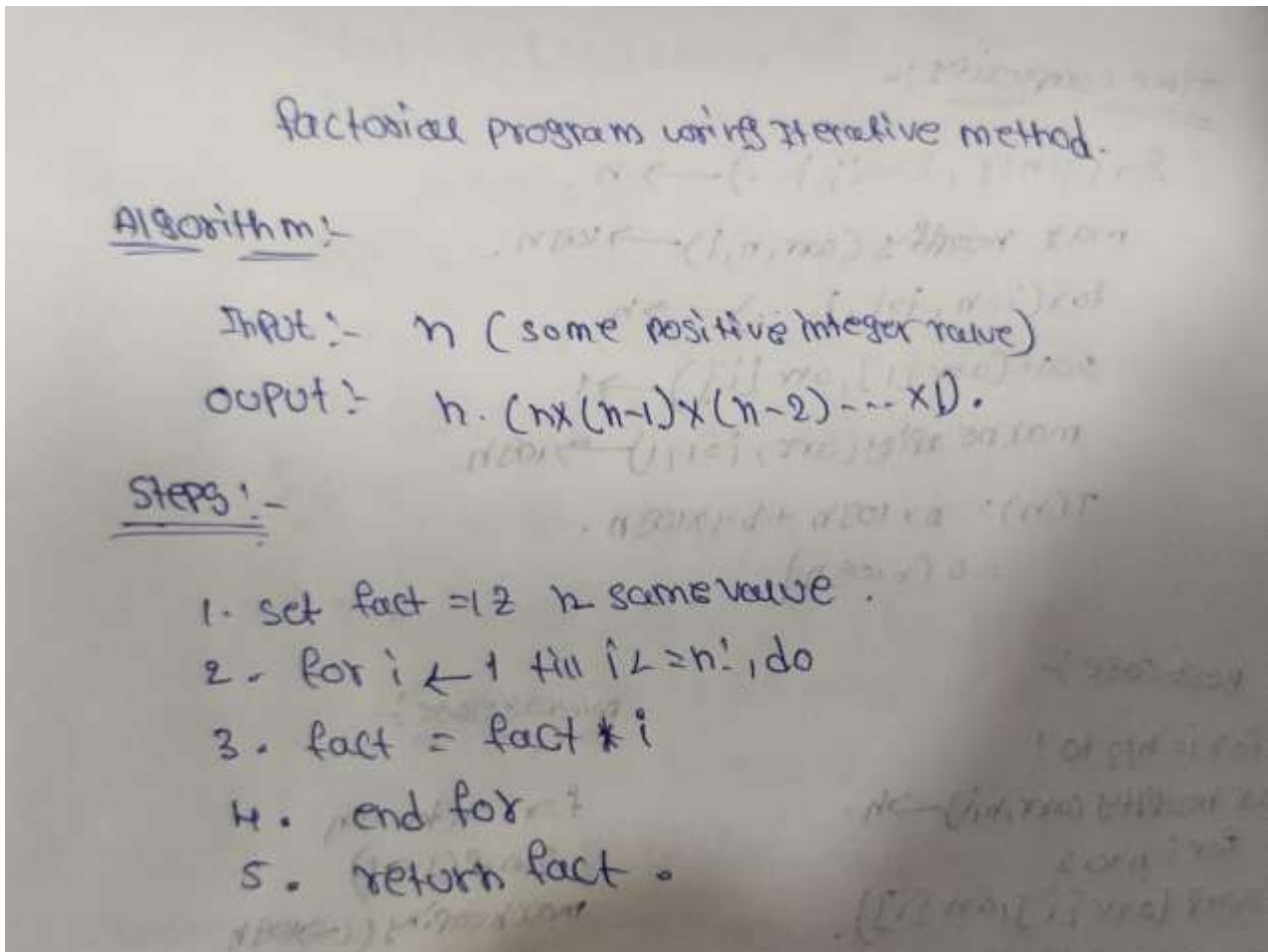


PRACTICAL-4

AIM: Implementation and Time analysis of factorial program using iterative and recursive method.

Factorial Using Iterative Method

ALGORITHM:



CODE:

```
#include <iostream>
using namespace std;
int factorial(int n)
{
    int fact = 1;
    for(int i = 1; i <= n; i++)
    {
        fact = fact * i;
    }
    return fact;
}
```

```
}  
  
int main()  
{  
    int n;  
    cout << "Enter a number: ";  
    cin >> n;  
  
    cout << "Factorial = " << factorial(n);  
  
    return 0;  
}
```

OUTPUT:

```
Output  
Enter a number: 6  
Factorial = 720  
  
=== Code Execution Successful ===
```

TIME COMPLEXITY:

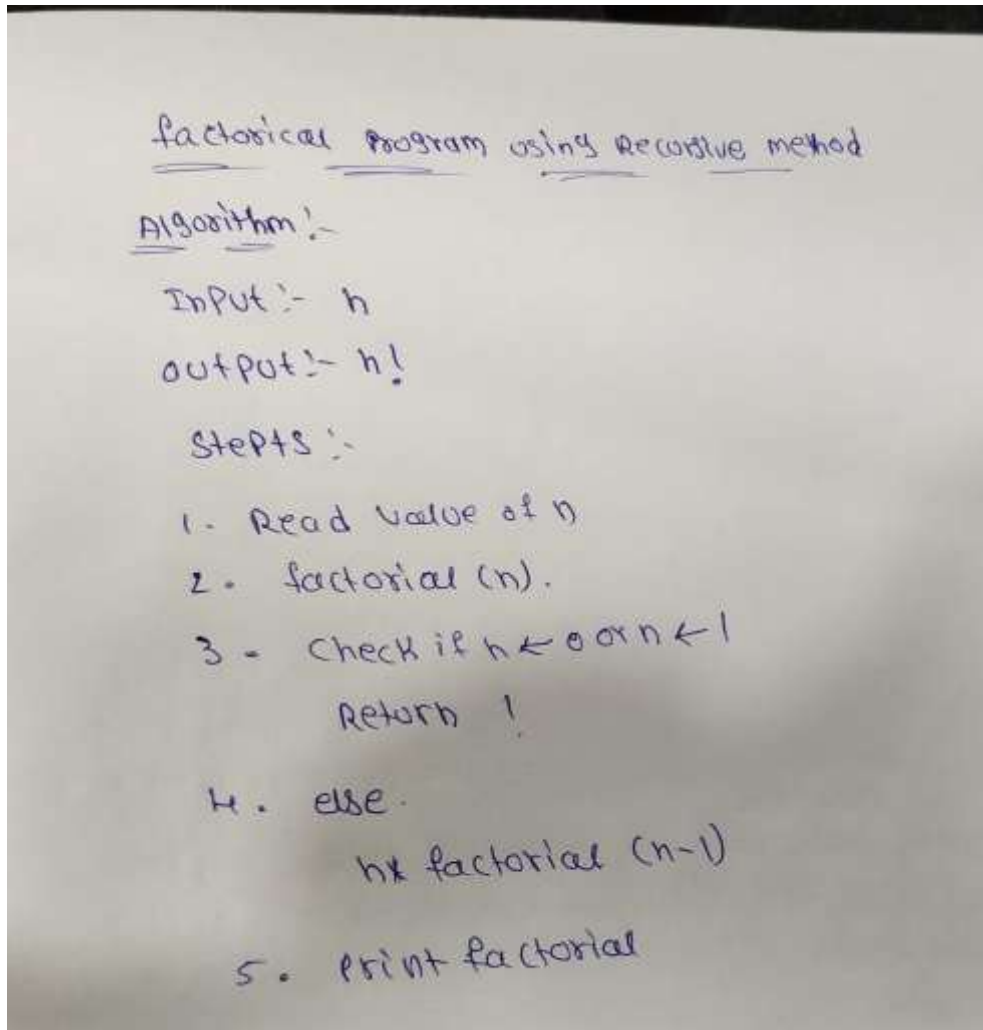
Time complexity

```
for (int i=1; i<=n; i++) {  
    fact = fact * i;  
}
```

$T(n) = n \times 1$
 $T(n) = O(n)$

Factorial Using Recursive Method:

Algorithm:

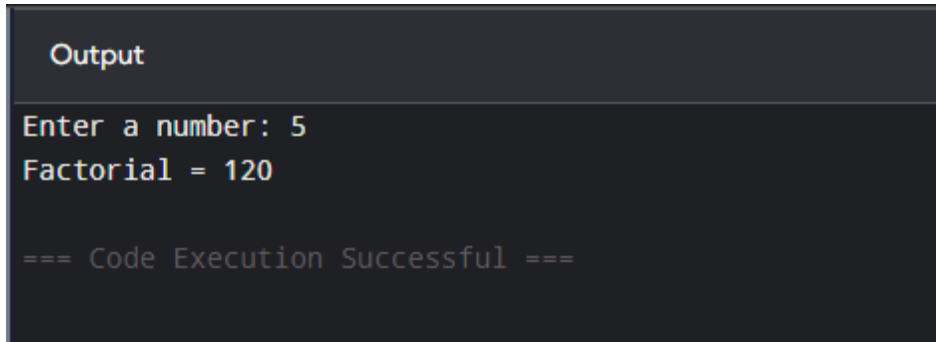


CODE:

```
#include <iostream>
using namespace std;
int factorial(int n)
{
    if(n == 0 || n == 1)
    {
        return 1;
    }
}
```

```
    return n * factorial(n - 1);  
}  
int main()  
{  
    int n;  
    cout << "Enter a number: ";  
    cin >> n;  
    cout << "Factorial = " << factorial(n);  
    return 0;  
}
```

OUTPUT:



The screenshot shows a dark-themed terminal window. At the top, the word "Output" is displayed in a light blue font. Below it, the program's execution is shown: "Enter a number: 5" followed by "Factorial = 120". At the bottom, a green message reads "=== Code Execution Successful ===".

TIME COMPLEXITY:

Time complexity:-

```
int factorial(int n) {  
    if (n == 0 || n == 1) {  $\rightarrow n$   
        return 1;  $\rightarrow 1$   
    }  
    else {  
        return n * factorial(n-1);  $\rightarrow n-1$   
    }  
}
```

$T(n) = \max(n, n-1).$
 $= n.$
 $T(n) = O(n),$

CONCLUSION:

The factorial program was successfully implemented using both iterative and recursive methods. Both methods produce the same result, but the recursive method uses function calls and stack space. The time complexity of both methods is $O(n)$, making them efficient for calculating factorials.